CHE 321 Chemical Reaction Engineering
EXAM 2, 100 points
Take-Home

Cheating includes looking at someone's exam paper (or allowing yours to be looked at), discussing the exam with someone, sharing/comparing answers, etc. Even asking 'what do you think about the paper?' is cheating. Any discussion about your paper with anyone before the due date is cheating, even if you do not copy. Looking for answers on the internet or any other source is not allowed. Any evidence of cheating will be thoroughly investigated. If anyone observes cheating/collaborative activities, it is your duty to inform the instructor.

Name ________________________________

1. (**15 points**) t-Butyl alcohol (TBA, C_{TBA}) is an important octane enhancer that is used to replace lead additives in gasoline. TBA was produced by the liquid-phase hydration (W, C_W) of isobutene (I, C_I) over an Amberlyst-15 catalyst. The system is normally a multiphase mixture of hydrocarbon, water, and solid catalysts. However, the use of co-solvents or excess TBA can achieve reasonable miscibility. The reaction mechanism is believed to be

   $I + S \leftrightarrow I{*}S$

   $W + I{*}S \leftrightarrow TBA{*}S$

   $TBA{*}S \leftrightarrow TBA + S$

   Derive the rate law assuming the Eley-Rideal kinetics is limiting.

2. (**50 points**) A first-order irreversible exothermic liquid-phase reaction:

   $$A \rightarrow B \qquad (k = 6.6 \times 10^{-3}\ \text{min}^{-1} \text{ at } 350\ \text{K};\ \ E = 40{,}000\ \text{cal/mol.K})$$

   is to be carried out in a jacketed CSTR (UA: 8000 cal/min.^{0}C, Ta = 300K). Species A and an inert I are fed to the reactor in equimolar amounts. The molar feed rate of A is 80 mol/min and τ = 100 min. The heat capacity of the A, B and I are 20, 20 and 30 cal/mol.^{0}C. The heat of reaction is 7500 cal/mol.

(a) Find the reactor temperature if the feed temperature is 450 K.
(b) Plot and then analyze the reactor temperature as a function of the feed temperature (from 310 K to 430 K).
(c) To what inlet temperature must the fluid be preheated for the reactor to operate at a high conversion? What are the corresponding temperature and conversion of the fluid in the CSTR at this inlet temperature?
(d) Suppose that the fluid inlet temperature is now heated 5 ^{0}C above the reactor temperature in part (c) and then cooled 20 ^{0}C, where it remains. What will be the conversion?
(e) What is the inlet extinction temperature for this reaction system?

3. (**35 points**) A second-order reaction is taking place in a micro-PBR reactor with micro-sized catalytic particles: A $\rightarrow$ B + 2C. The particle radius is 2 micrometers. A enters the micro-PBR with a superficial velocity is 3 micrometer/second at 100 ^{0}C and 5 atm. The reaction rate constant is found to be 10^{-4} m^6/mol.g.cat.s. Calculate the length of the micro-PBR to obtain a conversion of 99%.
   Diffusivity = 2.66 X 10^{-9} m^2/s
   Porosity of the micro-PBR = 0.4
   Catalyst's density = 100 g/m^3
   Internal Surface Area = 0.05 m^2/g

$$r_{AD} = k_{AD}\left(C_I C_V - C_{IS}/K_{IS}\right)$$

$$r_S = k_S\left(C_W C_{IS} - C_{TBA\cdot S}/K_S\right) \rightarrow \text{rate limited.}$$

$$r_D = k_D\left(C_{TBA\cdot S} - K_{TBA} C_{TBA} C_V\right)$$

$$r_{AD}/k_{AD} \approx r_D/k_D \approx 0 \Rightarrow C_{IS} = K_{IS} C_I C_V$$

$$C_{TBA\cdot S} = K_{TBA} C_{TBA} C_V.$$

Site Balance: $C_T = C_V + C_{IS} + C_{TBA\cdot S}$

$$= C_V + K_{IS} C_I C_V + K_{TBA} C_{TBA} C_V$$

$$C_V = C_T / (1 + K_{IS} C_I + K_{TBA} C_{TBA})$$

$$-r_I = r_S = \frac{k_S\left(K_{IS} C_I C_W - \frac{K_{TBA} C_{TBA}}{K_S}\right) C_T}{\left(1 + K_{IS} C_I + K_{TBA} C_{TBA}\right)}$$

$$= \frac{k\left(C_I C_W - C_{TBA}/K_H\right)}{1 + K_{IS} C_I + K_{TBA} C_{TBA}} \qquad k = k_S C_T K_{IS}$$

$$K_H = \frac{K_S K_{IS}}{K_{TBA}}$$

a.

$$G(T) = \Delta H_R X$$

$$X = \frac{\tau k}{1+\tau k}, k = 6.6\times10^{-3}\exp\left[\frac{E}{R}\left(\frac{1}{350} - \frac{1}{T}\right)\right]$$

$$R(T) = C_{p0}(1+\kappa)(T - T_c)$$

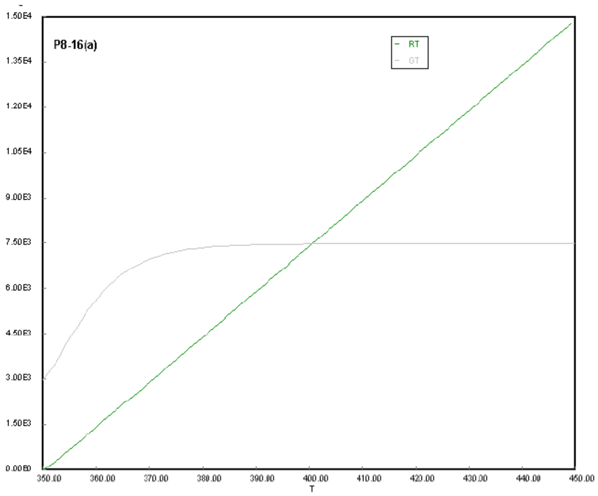
$$C_{p0} = \sum \Theta_i C_{pi} = 50$$

$$\kappa = \frac{UA}{C_{p0}F_{A0}} = \frac{8000}{50\times 80} = 2$$

$$Tc = \frac{\kappa T_a + T_0}{1+\kappa} = 350K$$

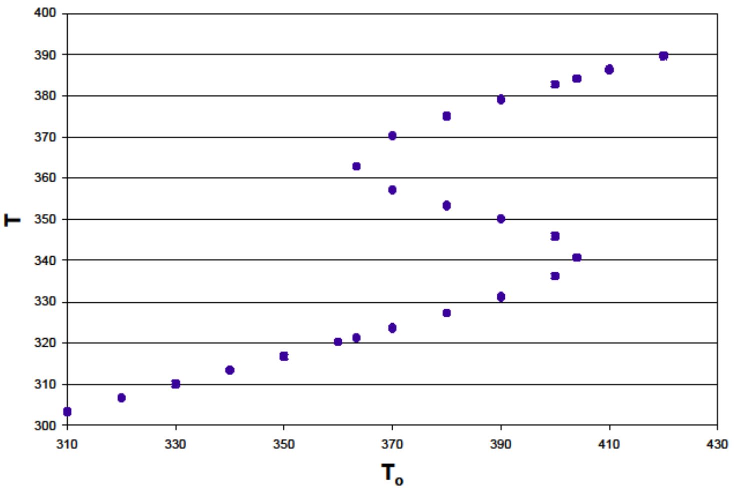
To find the steady state T, we must set G(T) = R(T). This can be done either graphically or solving the equations. We find that for T_0 = 450 K, steady state temperature 399.94 K.

Graphical method



b.

First, we must plot G(T) and R(T) for many different T_0's on the same plot. From this we must generate data that we use to plot T_s vs T_o.



(c) For high conversion, the feed stream must be pre-heated to at least 404 K. At this temperature, X = .991 and T = 384.2 K in the CSTR. Any feed temperature above this point will provide for higher conversions.

(d) For a temperature of 369.2 K, the conversion is 0.935

(e) The extinction temperature is 363.3 K (90oC).

3.

Rate law: $-r_A' = k\,C_A^2$

Mole balance:

$$D_{AB}\frac{d^2 C_A}{dz^2} - U\frac{dC_A}{dz} + r_A'\,\rho_B = 0$$

$$D_{AB}\frac{d^2 C_A}{dz^2} - U\frac{dC_A}{dz} - \Omega\,k\,S_A\,\rho_B\,C_A^2 = 0$$

Neglecting axial diffusion with respect to forced axial convection, we have:

$$\frac{dC_A}{dz} = -\left(\frac{\Omega\,k\,S_A\,\rho_B}{U}\right)C_A^2$$

$$-\int_{C_{A0}}^{C_A}\frac{dC_A}{C_A^2} = \left(\frac{\Omega\,k\,S_A\,\rho_B}{U}\right)\int_0^z dz$$

$$\frac{1}{C_A} - \frac{1}{C_{A0}} = \left(\frac{\Omega\,k\,S_A\,\rho_B}{U}\right)z$$

At $z = L$: $\quad \frac{1}{C_{A0}} = \left(\frac{1}{1-X} - 1\right) = \left(\frac{-2\rho_B\,k\,S_A}{U}\right)L$

$$L = \left(\frac{U}{\Omega\,\rho_B\,k\,S_A\,C_{A0}}\right)\left(\frac{1}{1-X} - 1\right)$$

Internal effective factor: $\quad \eta = \left(\frac{2}{n+1}\right)^{1/2}\frac{3}{\phi_n}$ where $n = 2$

$$\phi_2 = R\sqrt{\frac{k\,S_A\,\rho_B\,C_{A0}}{D_e}}$$

CA0 = P/RT

L =	**181.7106**	**micrometer**
U	3.00E-06	m/s
rho-B	1000	g/m3
k	0.001	m6/g/s/mol
SA	0.05	m2/g
CAO	0.032695	g/mol
X	0.99	
eta	**1.00E+00**	
Theile	**4.96E-02**	
De	2.66E-09	m2/s
T	373	K
P	1	atm
R	2.00E-06	m